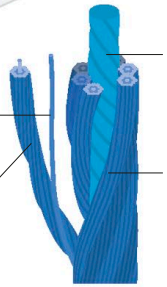
HOIST CABLE



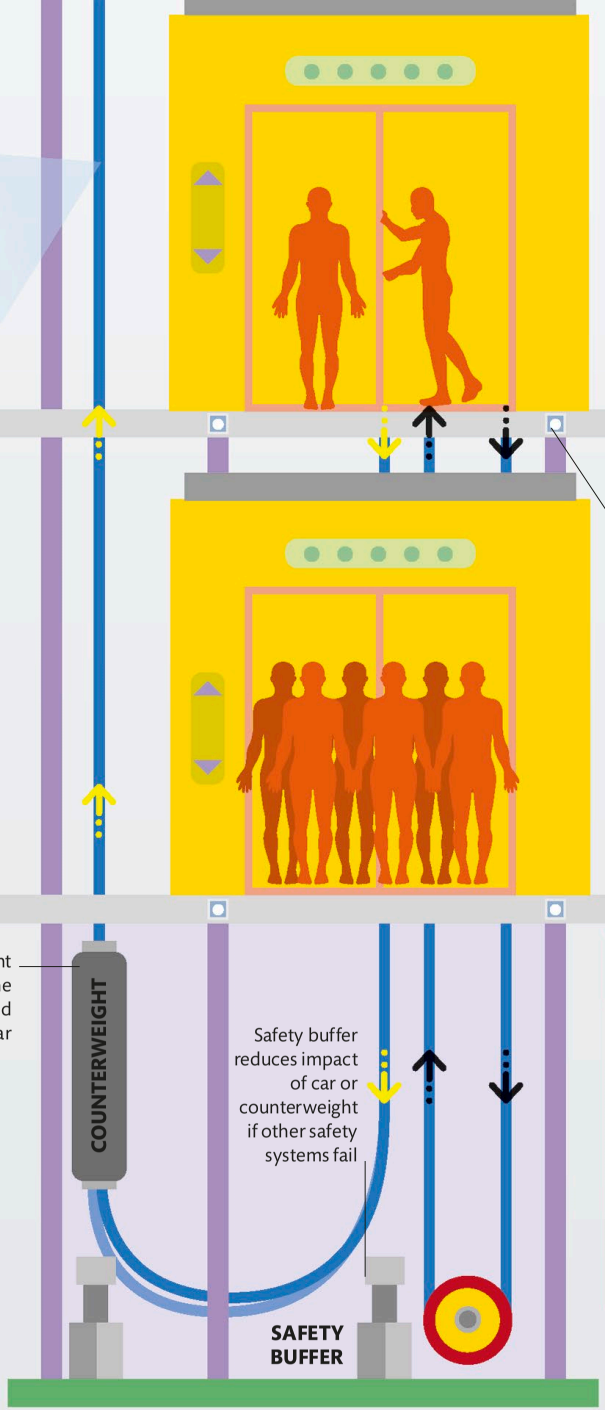
Each cable is made from many thin wires braided together. One cable can support the weight of the elevator car on its own, but most elevators have between four and eight cables.

ELEVATORS ARE THE SAFEST FORM OF TRAVEL AND ARE 50 TIMES SAFER THAN STAIRS



HOW FAST CAN AN ELEVATOR TRAVEL?

The fastest elevators can travel upward at about 67 ft (20.5 m) per second. The maximum down speed is about 33 ft (10 m) per second for most elevators.



Safety doors
Elevators have inner and outer doors. The inner doors are part of the elevator car, whereas the outer doors are part of the elevator shaft. The cars have a mechanism that unlocks the outer doors and pulls them open. In this way, the outer doors will open only if there is a car at that floor.

Sensors on guide rails detect whether elevator car is perfectly aligned with floor

Weight limit
All elevators have a maximum weight limit, which varies according to the size of elevator and its machinery. If the elevator's sensors detect overloading, they will prevent the doors from closing. Freight elevators are designed to carry much heavier loads than passenger elevators.

ELEVATOR PROGRAMMING

Elevators are controlled by computers that are programmed with efficient strategies for running the elevator cars. Usually, a car on its way up will not answer a "down" call until it has fulfilled all its "up" calls, and vice versa. Advanced systems take passenger traffic patterns into account and direct the elevator cars according to demand.